

SIMRAN KAUR

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RESEARCH INTERESTS

Large language model post-training, including self-improvement, skill composition, synthetic data, LLM evaluation (especially for creativity and compositional generalization), reinforcement learning, and instruction tuning; empirical foundations of deep learning.

EDUCATION

Princeton University Ph.D. in Computer Science · Advisor: Sanjeev Arora	Princeton, NJ 2022 – Present
Carnegie Mellon University B.S. Artificial Intelligence, School of Computer Science · GPA: 3.95 / 4.00	Pittsburgh, PA 2018 – 2022

RESEARCH EXPERIENCE

Princeton University , Graduate Researcher with Prof. Sanjeev Arora • Develop methods for LLM post-training under scarce supervision, including skill-mix instruction tuning (ICLR 2025), RL-induced skill composition (NeurIPS workshops 2025), and on-policy self-distillation (under review).	2022 – Present
Abridge , Research Intern • Fine-tuned BERT-based models and developed embedding methods to tag patient-clinician conversations with SOAP note-relevant labels, including review of systems, medications, and other clinical categories.	May 2021 – August 2021
Carnegie Mellon University , Undergraduate Researcher with Prof. Zachary C. Lipton • Studied empirical foundations of deep learning, including implicit regularization, optimization dynamics, and adversarial robustness.	2019 – 2022

PUBLICATIONS

* indicates equal contribution.

- [1] **Rethinking On-Policy Self-Distillation for Thinking Models** [\[Link\]](#)
Simran Kaur, Narutatsu Ri, Yinghui He, Liam Fowl, Sanjeev Arora.
FoGen Workshop (ICML), 2026.
- [2] **Self-Distillation Zero: Self-Revision Turns Binary Rewards into Dense Supervision** [\[Link\]](#)
Yinghui He, Simran Kaur, Adithya Bhaskar, Yongjin Yang, Jiarui Liu, Narutatsu Ri, Liam Fowl, Abhishek Panigrahi, Danqi Chen, Sanjeev Arora.
Oral at *RLxF Workshop* (ICML), 2026.
- [3] **How does RL Post-Training Induce Skill Composition? A Case Study on Countdown** [\[Link\]](#)
Simon Park*, Simran Kaur*, Sanjeev Arora.
Spotlight at *Efficient Reasoning Workshop* (NeurIPS), 2025; *MATH-AI Workshop* (NeurIPS), 2025.
- [4] **Instruct-SkillMix: A Powerful Pipeline for LLM Instruction Tuning** [\[Link\]](#)
Simran Kaur*, Simon Park*, Anirudh Goyal, Sanjeev Arora.
ICLR 2025. Oral at *Compositional Learning Workshop* (NeurIPS), 2024.
- [5] **Can Models Learn Skill Composition from Examples?** [\[Link\]](#)
Haoyu Zhao, Simran Kaur, Dingli Yu, Anirudh Goyal, Sanjeev Arora.
NeurIPS 2024.
- [6] **Skill-Mix: a Flexible and Expandable Family of Evaluations for AI models** [\[Link\]](#) [\[Quanta Article\]](#)
Dingli Yu, Simran Kaur, Arushi Gupta, Jonah Brown-Cohen, Anirudh Goyal, Sanjeev Arora.
ICLR 2024; *Workshop on Distribution Shifts* (NeurIPS), 2023.
- [7] **Disentangling the Mechanisms behind Implicit Regularization in SGD** [\[Link\]](#)
Zachary Novack, Simran Kaur, Tanya Marwah, Saurabh Garg, Zachary C. Lipton.

ICLR 2023. Spotlight at *Higher Order Optimization in ML Workshop* (NeurIPS), 2022. [Best Poster Award]

[8] ***On the Maximum Hessian Eigenvalue and Generalization*** [\[Link\]](#)

Simran Kaur, Jeremy Cohen, Zachary C. Lipton.

Contributed talk at "*I Can't Believe It's Not Better!*" Workshop (NeurIPS), 2022.

[9] ***Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability*** [\[Link\]](#)

Jeremy M. Cohen, Simran Kaur, Yuanzhi Li, Zico Kolter, Ameet Talwalkar.

ICLR 2021; *Opt2020 Workshop* (NeurIPS), 2020.

[10] ***Are Perceptually-Aligned Gradients a General Property of Robust Classifiers?*** [\[Link\]](#)

Simran Kaur, Jeremy Cohen, Zachary C. Lipton.

Science Meets Engineering of Deep Learning Workshop (NeurIPS), 2019.

TALKS & PRESENTATIONS

<i>How does RL Induce Skill Composition?</i> , Princeton Language and Intelligence (PLI)	October 2025
<i>Instruct-SkillMix</i> , Princeton Language and Intelligence (PLI)	October 2024
<i>Instruct-SkillMix</i> , Compositional Learning Workshop, NeurIPS	December 2024
<i>Instruct-SkillMix</i> , Danqi Chen Lab Meeting	June 2024
<i>On the Maximum Hessian Eigenvalue and Generalization</i> , ICBINB Workshop, NeurIPS	December 2022
<i>On the Maximum Hessian Eigenvalue and Generalization</i> , Elad Hazan Lab Meeting	October 2022

HONORS & AWARDS

Princeton AI Lab Graduate Student Fellow	September 2025 – Present
Best Poster Award , Higher Order Optimization in ML Workshop (NeurIPS)	2022
CMU Senior Leadership Recognition Recipient	May 2022
Phi Beta Kappa	May 2022
CMU SCS College Honors (senior thesis)	May 2022
Dean's List	Spring 2019 – Spring 2022

TEACHING

Princeton University: Teaching Assistant, <i>COS 324: Introduction to ML</i>	Fall 2023, Spring 2024
CMU: Head TA, <i>15-281: AI: Representation and Problem Solving</i>	Spring 2021 – Spring 2022
CMU: TA, <i>15-281: AI: Representation and Problem Solving</i>	Spring 2020, Fall 2020
CMU: TA, <i>10-301/10-601: Introduction to ML</i>	Summer 2020

SERVICE

Conference Reviewing: NeurIPS, ICLR	2023 – Present
Co-organizer, Princeton Algorithms & Machine Learning (Alg-ML) Seminar	Fall 2023 – Spring 2024
Princeton Language and Intelligence (PLI) Blog	Fall 2023 – Present

TECHNICAL SKILLS

Languages: Python, C, Standard ML, R

ML & Training: PyTorch, Hugging Face (Transformers, TRL, Datasets), verl, DeepSpeed, FSDP, Accelerate

Inference & Tooling: vLLM, lm-evaluation-harness, Slurm, Weights & Biases, Git, Jupyter